

Morphologic and Functional Changes of Macula and Optic Nerve Head in Adult Kidney Transplant Recipient

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ABSTRACT

Aim. To assess macular and optic nerve head characteristics in patients after kidney transplantation using noninvasive optical coherence tomography (OCT) imaging.

Methods. The study had a prospective character and was performed during period from November 2019 to November 2020 at the University Clinical Center Tuzla, Tuzla, Bosnia and Herzegovina, and included 59 patients with renal graft who underwent OCT. The obtained OCT images, pathologic changes on the fundus, including macula and optic nerve, were analyzed by reviewing nonparametric measures such as scanning laser ophthalmoscope, B-scan, 3D display, as well as parametric measurements such as central macular thickness, optic nerve head excavation, and retinal nerve thickness fiber (RNFL).

Results. There is a statistically significant difference in RNFL values ($P = 0.031$) and cup to disc ($P = 0.002$) ratio in relation to duration of graft. The highest average values of RNFL was in patients with creatinine clearance <30 mL/min, while average cup to disc ratio was lowest in patients with creatinine clearance >60 mL/min.

Optic nerve head edema was bilateral in 9 and unilateral in 5 patients. Fundus examination also revealed changes on posterior segment, including retinal detachment (1 patient), bilateral macular edema (9 patients), unilateral macular edema (3 patients), central serous chorioretinopathy (2 patients), and macular pseudo hole (3 patients).

Conclusions. Different retinal pathology is present in high incidence in patients secondary to renal transplant. Immunosuppressive therapy, older age, and comorbidities are the usual causes of these changes.

PATHOLOGIC ophthalmologic findings in organ transplant patients occur in more than 50% of patients [1], seldom causing significant visual loss with a need for an urgent treatment (3% of cases) [2]. Chronicity of disease, underlying damaging process, and immunosuppressive therapy are usual causes of these changes [3]. The most common ophthalmologic changes secondary to organ transplantation are corticosteroid-induced cataract and open-angle glaucoma [3]. Posterior segment complications have been described for bone marrow and heart transplantation [3], but little is known about kidney-transplanted patients, which is the most common transplanted organ worldwide. Organ transplantation cannot be performed without high doses of immunosuppressive therapy, such as steroids, calcineurin inhibitors, cyclosporine, tacrolimus, proliferation signal inhibitors, and inhibitors of cell division [4]. All these

medications come with side effects, often affecting photoreceptors and optic nerve [4]. Ophthalmologic changes after kidney transplant are usually a progression of previous ophthalmologic status [4]. Preoperative ophthalmologic examination should include best corrected visual acuity, intraocular tonometry, anterior segment examination, dilated posterior segment examination, and optical coherence tomography (OCT). Named tests are noninvasive and simple procedures that can be easily performed and monitored. OCT allows a great insight into posterior segment pathology, including macula, and optic nerve head

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(ONH) providing high-resolution, cross-sectional images. Kidney disease is usually accompanied by hypertension and diabetes and is accompanied by endothelial dysfunction, while endothelium plays an important role in vascular biology and regulation of renal function [5]. Dysfunctional endothelium is characterized by reduced bioavailability of nitric oxide and excessive production of endothelin-1, which disrupts vascular hemostasis, increases expression of adhesion molecules, increases blood thrombogenicity, increases expression of cellular endothelial adhesion molecules, and leads to adhesion of inflammatory cells [5]. The result is disruption of the function of tissues or organs. Diabetic retinopathy (DR) should always be evaluated preoperatively since it can sometimes develop as a consequence of new-onset diabetes after transplantation [4]. Proliferative DR (PDR) has uncertain prognosis and should be monitored and treated more carefully. Kidney transplantation is considered to stabilize retinopathy (both diabetic and hypertensive) as well as diabetic macular oedema (MO) [6]. The risk of central retinal artery occlusion is also higher in patients with chronic kidney disease [7]. Some studies show a decrease in retinal nerve fiber layer (RNFL) thickness, which indicates neuropathy progression [8]. Available studies investigate anterior segment changes after organ transplants due to immunosuppressive therapy, but not many studies have yet reported the effects of kidney transplants on the posterior segment of the eye. It is also the first such study in Bosnia and Herzegovina. The aim of this study is to assess macular and ONH characteristics in patients after kidney transplantation using noninvasive OCT imaging.

MATERIALS AND METHODS

According to the duration of the graft, the groups were divided into group I from the first to the end of the fifth year, group II from 6th to the end of the 10th year, and group III over 10 years of the graft duration. In all patients, the functional state of the graft was determined according to the estimated creatinine clearance by Cockcroft-Gault equation. According to results, sample was divided into: (A) creatinine clearance above 60 mL/min, (B) creatinine clearance of 30 to 60 mL/min, and (C) creatinine clearance 15 to 30 mL/min. The following demographics were taken from all patients: sex, age, body weight,

height, date, and type of transplantation. After obtaining informed consent, ophthalmologic examination was performed and included dilated posterior segment examination with OCT imaging using Zeiss Stratus and Optopol Revo NX OCT. The obtained OCT images, pathologic changes on the fundus, including macula and optic nerve, were analyzed by reviewing nonparametric measures such as scanning laser ophthalmoscope, B-scan, 3D display, as well as parametric measurements such as central macular thickness, optic nerve head excavation (cup to disc ratio (C/D)), and RNFL. Criteria for inclusion were data on kidney transplantation (living related and living unrelated donor). Exclusion criteria were patients who were not able to undergo the examination due to various reasons, patients with opaque media due to corneal opacities, advanced cataract, or vitreous haemorrhage which disables OCT recording, and patients post vitrectomy surgery or anti-VEGF application. The research was approved by the Ethics Committee of the University Clinical Center Tuzla.

Data processing was done by standard procedures of descriptive and inferential statistics. The corresponding tables give the minimum and maximum values of the tested quantities as well as their mean values and the corresponding values of the standard deviation. The paper compares the values of the examined values in 3 groups, so the testing of the statistical significance of the differences in the mean values was done by the Analysis of Variance procedure. Post hoc testing (post testing) after testing by the Analysis of Variance procedure, in order to determine the statistical significance of the difference between the mean values between the groups individually, was done by the Scheffé test. Statistical hypotheses were tested with a significance level of 5% (0.05).

RESULTS

Descriptive analysis of age, dialysis time, body mass index, and creatinine clearance are shown in Table 1. Patients were divided in relation to the duration of the renal graft, and OCT analysis was performed (Table 2). There was no statistically significant difference between the mean values of the observed parameters in the observed groups ($P > .05$). Analysis of OCT parameters in relation to creatinine clearance values is shown in Table 3. There is a statistically significant difference in RNFL values ($P = .031$) and C/D ($P = .002$) ratio between groups, while the same was not verified for CFT ($P = .251$). In order to determine between which groups there are statistically significant differences in the mean values, a post hoc (subsequent) analysis was performed

Table 1. Descriptive Sample Representation

Sex		N	Min	Max	Mean	SD
Male	Age (y)	31	23	63	46.65	10.87
	Dialysis time (mo)	30	4	216	57.70	56.99
	BMI (kg/m ²)	30	15.80	30.50	24.85	3.31
	Creatinine clearance	31	23	115	59.19	20.00
Female	Age (y)	28	19	76	51.04	12.61
	Dialysis time (mo)	27	0	132	49.37	39.95
	BMI (kg/m ²)	26	17.60	43.00	26.87	5.26
	Creatinine clearance	27	22	112	59.22	22.30
Total	Age (y)	59	19	76	48.73	11.833
	Dialysis time (mo)	57	0	216	53.75	49.40
	BMI (kg/m ²)	56	15.80	43.00	25.79	4.40
	Creatinine clearance	59	22	115	59.21	20.91

BMI, body mass index; eGFR, estimated glomerular filtration rate; SD, standard deviation.

Table 2. OCT Parameters in Relation to Renal Graft Life

Graft age in y	Parameter	N	Min	Max	Mean	SD
≤5	RNFL (microns)	41	88.00	125.00	102.73	8.48
	C/D	34	0.01	0.70	0.36	0.18
	CFT (microns)	44	147.000	246.000	192.614	21.43
5-10	RNFL (microns)	51	40.960	131.000	103.181	16.954
	C/D	41	0.060	0.877	0.390	0.200
	CFT (microns)	50	122.000	435.000	203.520	57.017
10+	RNFL (microns)	22	85.000	228.000	109.700	29.683
	C/D	17	0.034	0.530	0.354	0.166
	CFT (microns)	22	141.000	413.000	218.545	65.210

C/D, cup to disc ratio; CFT, central foveal thickness; RNFL, retinal nerve fibre layer; SD, standard deviation; X, mean value.

Table 3. OCT Parameters in Relation to Creatinine Clearance

Creatinine clearance		N	Min	Max	Mean	SD
>60 mL/min	RNFL (microns)	41	65.000	129.000	102.147	11.938
	C/D	32	0.010	0.690	0.285	0.149
	CFT (microns)	42	147.000	435.000	208.809	60.599
30-60 mL/min	RNFL (microns)	63	40.960	131.000	103.429	14.290
	C/D	55	0.042	0.877	0.426	0.189
	CFT (microns)	64	122.000	293.000	195.641	34.230
<30 mL/min	RNFL (microns)	10	88.000	228.000	118.354	42.141
	C/D	5	0.100	0.670	0.411	0.203
	CFT (microns)	10	151.000	413.000	216.800	72.464

C/D, cup to disc ratio; CFT, central foveal thickness; eCrCl, creatinine clearance; RNFL, retinal nerve fibre layer; SD, standard deviation; X, mean value.

Table 4. Results of Post Hoc Testing of the Statistical Significance of the Difference in Mean Values of Changes in OCT Parameters Between Groups Individually

ONH OCT parameter	(I) e Creatinine clearance	(J) e Creatinine clearance	Difference (I-J)	P value
RNFL	>60 mL/min	(30-60) mL/min	-1.28	.936
	>60 mL/min	<30 mL/min	-16.21	.036
	(30-60) mL/min	<30 mL/min	-14.92	.049
	>60 mL/min	(30-60) mL/min	-0.14	.002
C/D	>60 mL/min	<30 mL/min	-0.13	.336
	(30-60) mL/min	<30 mL/min	0.02	.983

C/D, cup to disc ratio; ONH, optic nerve head; P value, level of significance.

using the Scheffe procedure. The results of the analysis are shown in Table 4. An interesting finding is ONH edema, which was bilateral in 9 and unilateral in 5 patients. Fundus examination also revealed changes on posterior segment, including bilateral MO (9 patients), unilateral MO (3 patients), central serous chorioretinopathy (CSC) (2 patients), and macular pseudo hole (3 patients) and retinal detachment (1 patient).

DISCUSSION

Studies show that there is a positive significant correlation between progression of renal failure and retinal changes [9]. It has also been proved that severity of these pathologic changes also increases with the progression of renal failure [10]. Observed ophthalmologic parameters were RNFL thickness, C/

D, and central foveal thickness (CFT) in relation to both duration of graft receiving and creatinine clearance. Average CFT in all patients was slightly higher than in healthy eyes ($182 \pm 23 \mu\text{m}$) [11]. It showed a gradual increase within the groups, with the highest CFT in patients who had received the graft more than 10 years ago and with creatinine clearance <30 mL/min. Patients with chronic kidney disease usually show reduction in CFT due to compromised vascularization due to vascular injury, leading to atrophy [12]. Higher average central thickness in our study can be due to different mechanisms. Firstly, it might be due to IS therapy post kidney transplantation. There is also a higher incidence of patients with MO, which increases average values of CFT. The third reason would be normalization of overall vasculature posttransplant, affecting retina as well. Balmaforth et al explained that decreased thickness was limited to

outer retinal layers receiving vascular supply from the choroid [12]. This can explain lack of association with CFT values in patients with low creatinine clearance. Average RNFL thickness in our study was within the values of normal eyes ($104.17 \pm 10.71 \mu\text{m}$ [13]) and had also showed the same trend of increase as in CFT. Minimum RNFL values are significantly decreased in all groups of patients, which is consistent with findings in other studies. Raczynska et al reported a decrease in RNFL thickness, suspecting that IS therapy has a toxic effect on oligodendrocytes and glial cells [4]. Berindán et al had also reported decreased RNFL in almost 75.8% of examined eyes [8]. Chronic kidney disease causes vascular ischemia that also affects retinal circulation, leading to RNFL decrease. Our study has shown the difference in RNFL thickness among groups according to eGFR values, with the highest average values of RNFL in patients with creatinine clearance $<30 \text{ mL/min}$, which is not comparable to other studies [4,8]. Normal average RNFL thickness in our study can be explained by unusually high incidence of ONH oedema that has increased average RNFL values. It can also be taken in consideration that kidney transplantation significantly improves patients' ophthalmologic status. Average C/D ratio was within physiological range and has shown the difference among the groups according to eGFR values with the lowest values in patients with creatinine clearance $>60 \text{ mL/min}$. It must be taken in consideration that C/D was not used in statistical analysis in patients with ONH edema due to unmeasurable values, which decreases average value significantly. Maximum values of C/D were increased in all 3 groups, which matches the decreased RNFL values due to vascular ischemia caused by renal failure. Unilateral ONH was found in 5 and bilateral in 9 patients from our study. Other studies reported the presence of optic neuropathy after iIS therapy in kidney transplant patients [4,14,15]. High incidence of ONH oedema in our study can be due to prolonged iIS therapy that causes axonal swelling, or the presence of underlying ophthalmologic disease, such as hypertensive retinopathy and DR. CSC was found in 2 eyes, MO in 12 eyes, and retinal detachment in 4 eyes. This proves the benefits of regular ophthalmologic examination in renal transplant recipients. Van Dijk et al reported higher incidence of CSC (27%) with macular subretinal fluid in 2 patients [16]. They also reported the presence of epiretinal membrane in 16% of patients and retinal detachment in 2 patients [17]. CSC, together with increased CFT, could be corticosteroid induced [16]. It is still not known whether retinal abnormalities are caused due to short-term high-dose or long-term low corticosteroid dose with cumulative effect. Even though all patients are on IS therapy postrenal transplantation, not all of them develop ophthalmologic changes. This means that more risk factors are probably included in retinal pathology. Kian-Ersi et al described the presence of CSC, retinal pigment epitheliopathy, retinal vein occlusion, and macular oedema in renal transplant patients [18]. The incidence of these findings is similar to our study. High incidence of MO and DR in these patients can be due to present comorbidity-related ophthalmologic condition, such as NPDR, PDR, or DME. Some changes can also be present due to advanced age, such as age-related macular degeneration or decrease in RNFL. Limitation of this study is inability to

analyze ophthalmologic changes over a period of time. Also, it has not been indicated whether the patients already had changes on posterior segment due to other comorbidities, such as HR and DR. The advantage of this study is sample size. The study can be improved by using OCT angiography in these patients because of shared anatomic and physiological circulation of kidneys and retina. Further improvement to this study could be made by including more parameters, such as visual acuity, fundus changes recorded on fundus camera, and visual field defects. The current pandemic disabled further investigation and more frequent follow-ups due to increased risk of infection in immunocompromised transplanted patients. This study can serve as a pioneer study for larger study, which could include more ophthalmologic measures performed before and after the transplant.

CONCLUSIONS

Different retinal pathology is present in high incidence in patients secondary to renal transplant. Immunosuppressive therapy, older age, and comorbidities are the usual cause of these changes. Regular ophthalmologic follow-ups with dilated posterior segment and OCT examination are therefore highly recommended.

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

DATA AVAILABILITY

Data will be made available on request.

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